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Conference 2000



Bluetooth

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Session Agenda

- What is Bluetooth?
- Usage scenarios
- Technical Overview
- Bluetooth status
- Palm’s Bluetooth hardware
- Bluetooth software
- Roadmap

What Is Bluetooth?

- Wireless link
  - Specification covers both hardware and software
  - Uses the open 2.4 GHz frequency range
  - This frequency is available worldwide
- Designed as a cable replacement
  - But it offers so much more…
- Analyst figures show over 1,000 million Bluetooth devices in 2005*

*Cahners In-Stat Group

Bluetooth Products

- Mobile phones
- Mobile PCs (laptops)
- Handheld devices
- Wireless peripherals
  - Headsets
  - Cameras
- Access Points
  - Bridge to wired networks

Why Will Bluetooth Succeed?

- Open standard
  - Specification freely available
  - www.bluetooth.com
- Very focused on mobile wireless links
  - Very small
  - Low cost
  - Low power consumption
- Huge industry support
  - Over 2,000 member companies!

Usage Scenarios

Ultimate Headset

- Same headset for work, home and on the move



Usage Scenarios

Car Integration

- Bluetooth built into car stereos
  - Use any cell phone in any car!



Palm Usage Scenarios

Palm and Cellphone

- Integral part of Palm's wireless strategy
- Surf on your Palm, using your cellphone in your backpack!



Palm Usage Scenarios

Palm to Palm

- Multi-player games
- Collaborative applications
- Long-distance “beaming”



Palm Usage Scenarios

Bluetooth in the Enterprise

- Create “hot spots” for information push
- Cradle-less data synchronization



Palm Usage Scenarios

Palm to Other Devices

- Read gas and electricity meters wirelessly
- Access control to buildings
- Unlock your car



Technical Features

- 2.4 GHz Open band
  - Globally available
  - Frequency hopping and Time Division Multiplexing
- 10–100 meter range (“personal bubble”)
  - Up to 8 active devices per bubble
- 1 Mbps gross rate
- Simultaneous voice / data capable
  - 432 Kbps (full duplex), 721/56 Kbps asymmetric
  - Or 3 simultaneous voice full duplex calls
  - Or a combination of voice and data

What Else Uses 2.4 GHz?

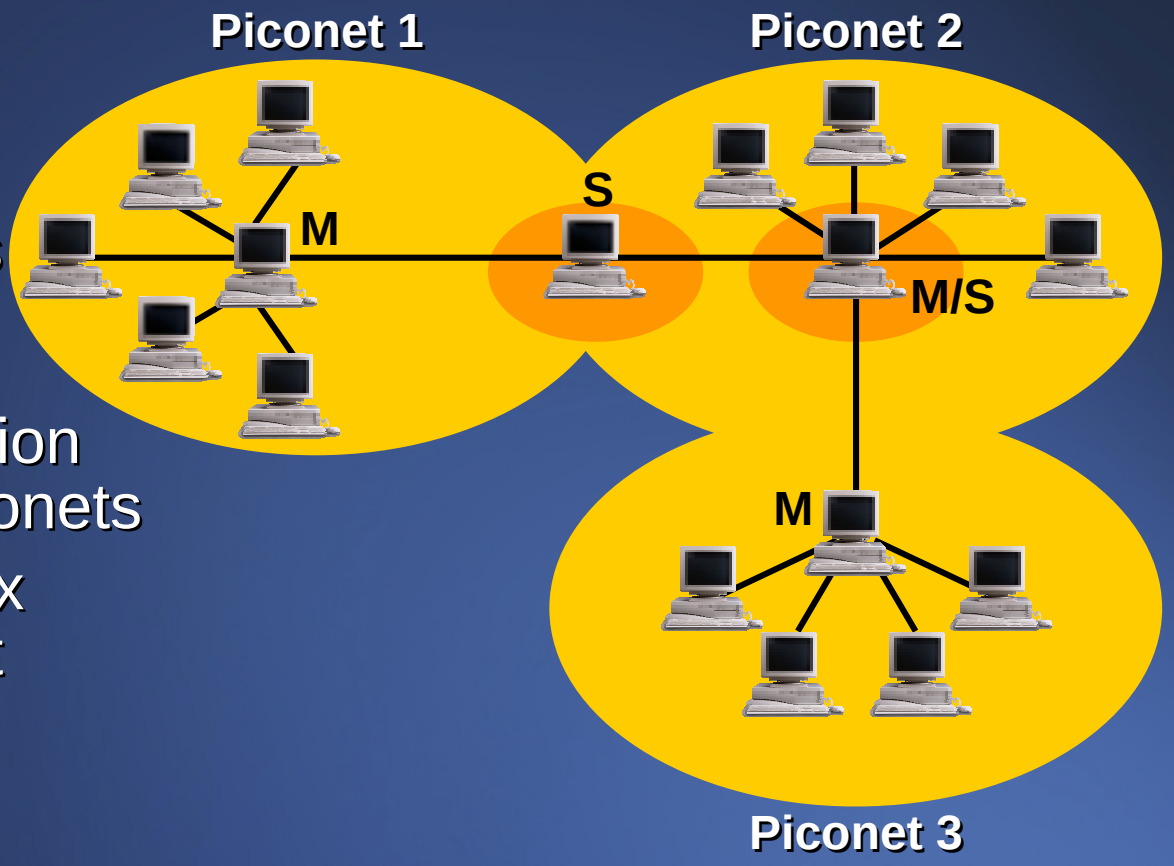
- Your microwave oven!!!
- Cordless phones
- 802.11B and HomeRF
  - These have higher bandwidth
  - But:
    - They don't support voice
    - They are relatively expensive
    - They require too much power
    - Bluetooth doesn't require a hub
- Bluetooth is robust

What Makes Bluetooth Robust?

- Frequency hops at 1600 times per second
  - Changes frequency every 625 μs
- Uses short and flexible packets
  - Packets are 1, 3 or 5 timeslots
- Very fast error detection and correction

Bluetooth Topology

- Piconet
  - 1 master, up to 7 active slaves
- Scatternet
  - Communication between Piconets
  - Very complex to implement
  - Won't be supported for a while



Bluetooth Status

Results from Unplugfest in November

- Earlier specification (1.0B) had problems
  - For example, flow control in RFCOMM
- Specification 1.1 fixed most of these
  - Spec was published in October
- Palm is one of the leaders
- Most vendors have point-to-point working
- Some vendors have point-to-multipoint working
- Scatternet will not happen this generation of radio

Palm's Bluetooth Hardware

- Palm's Bluetooth initiative started by 3Com over 2 years ago
- Bluetooth will be an add-on until prices drop
  - Bluetooth will eventually be integrated into Palms
- Power 1mW (0 dBm)
  - 10 meter (30 feet) range between Palms
  - 30 meters (100 feet) to LAN Access Point
- Uses serial port
  - Speed limited to 115,200 bps

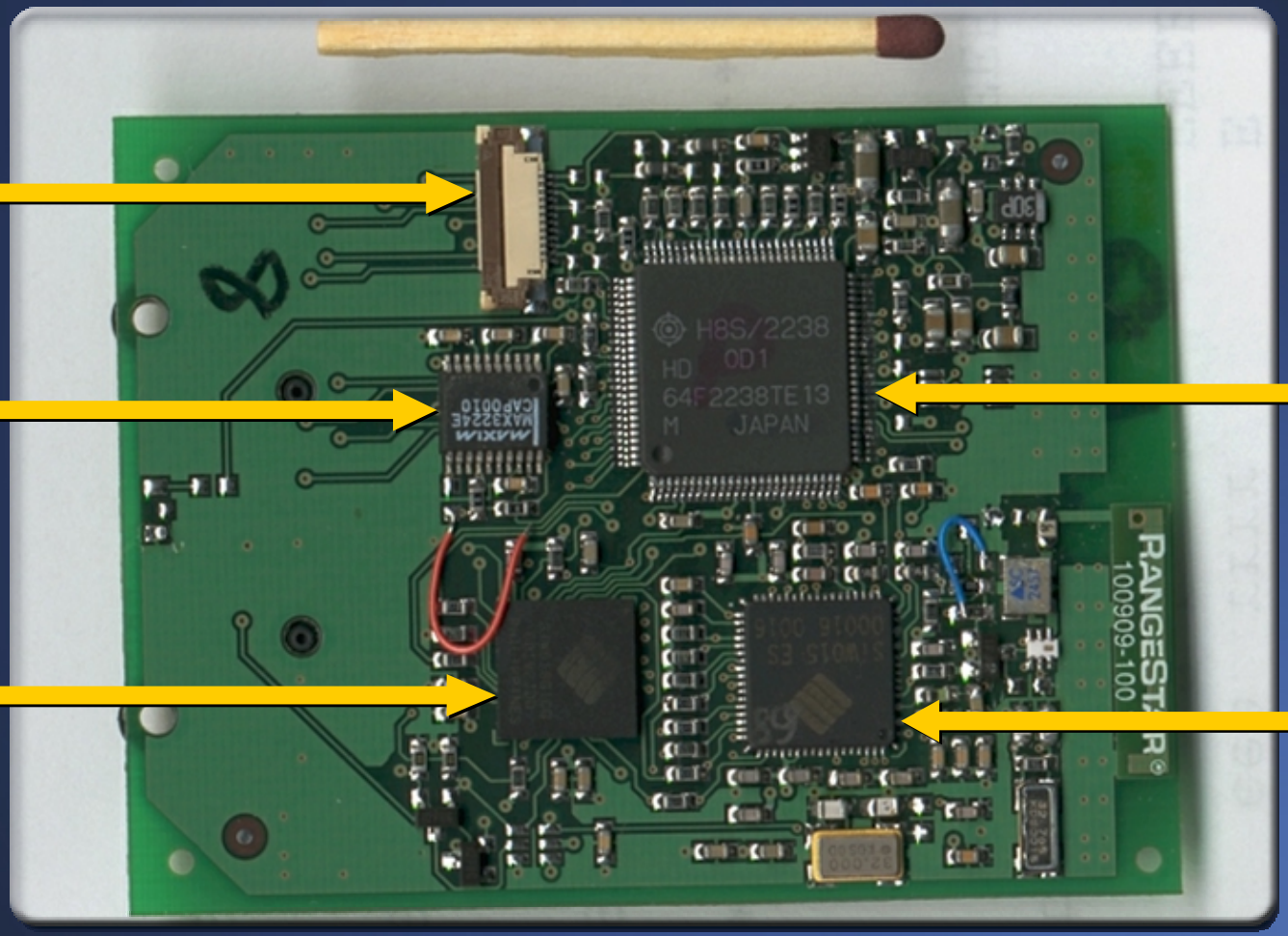
Palm Vx Bluetooth Hardcase



Palm Vx Bluetooth Hardcase



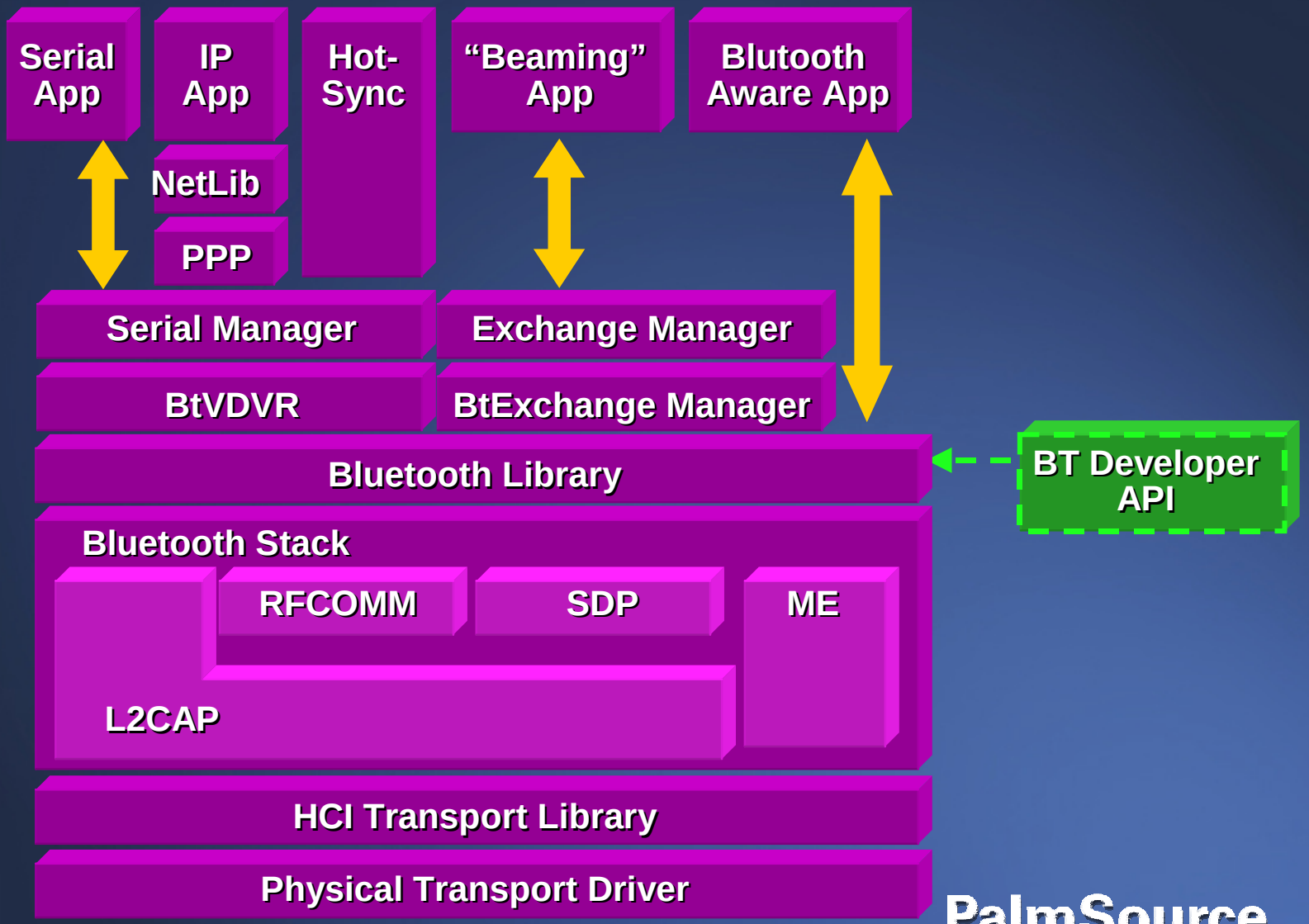
Bluetooth Radio Module



Hardware Operation

- The Palm Vx powers the Bluetooth module
- Bluetooth module monitors Tx and RTS pin
  - Goes to standby mode when lines not powered
- Bluetooth module asserts HotSync interrupt to wakeup Palm
  - HotSync notification send to Bluetooth library

Software Stack



Bluetooth Connections

- Voice or data connections
  - Asynchronous Connectionless (ACL)
    - Used for data
  - Synchronous Connection-Oriented (SCO)
    - Used for voice
    - Not supported in this release

- Paired or temporary connections
  - Temporary connections have to be “discovered”
  - Paired connections have benefits:
    - Security
    - Faster connections

Typical Bluetooth Operation

- Discover single or multiple devices
  - Average time to connect:
    - Paired = 0.64 seconds
    - Temporary = 2.56 seconds
- Create an ACL (data) link to the device
- Create a socket:
  - RFCOMM (Used by Virtual Serial Driver)
  - L2CAP = Data connection
  - SDP = Service Discovery Protocol
- Pass data
- Close socket, close ACL link

Software

- Bluetooth will not be released at the same time as OS 4.0
  - However, OS 4.0 will be “Bluetooth-aware”
- Bluetooth libraries will be released after OS 4.0
  - OS revision number won’t be increased
    - “OS 4.0 with Bluetooth”
  - Runs on POSE
  - Around 200K

Product Roadmap

- Hardcase for Palm Vx
  - Seed some developers in Jan / Feb
    - Software and hardware will be alpha
  - General developer availability Q2 2001
  - This is the “reference design”
- Hardcase for other Palms?
- Bluetooth on SD card?
- Bluetooth integrated into Palms
  - Will happen, probably 2002 or 2003
- Licensees

Developer Resources

- PalmSource
  - Two Bluetooth labs
    - Wed 3:45–4:45pm (Large Lab)
    - Fri 10–11am (Large Lab)
  - Open labs
    - 6:45pm–1am
- Provider Pavilion (www.palmos.com)
  - White Paper very soon!
  - Bluetooth APIs


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